EXPERIMENT 05

Word Stemming using Porter Stemmer Algorithm

AIM

To perform word stemming on a list of words using the Porter Stemmer algorithm via Python's NLTK library.

PROCEDURE

1. Import `PorterStemmer` from `nltk.stem`.
2. Instantiate the PorterStemmer object.
3. Create a list of words exhibiting varying inflections, derivations, and morphological suffixes.
4. Iterate through the list and apply the stemmer's `stem()` method to each word.
5. Display the original word alongside its resulting stem and analyze affix reductions.

PROGRAM

```
import nltk
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

word_list = [
    "connect", "connection", "connected", "connecting", "connective",
    "program", "programs", "programmer", "programming", "programmed",
    "compute", "computation", "computing", "computer", "computable",
    "generous", "generosity", "revival", "allowance", "rational"
]

print(f"{'Original Word':<16} | {'Stemmed Output':<15} | {'Suffix Stripped'}")
print("-" * 50)

for word in word_list:
    stem = stemmer.stem(word)
    stripped = word[len(stem):] if word.startswith(stem) else "Non-linear /
Modified"
    print(f"{word:<16} | {stem:<15} | {stripped}")
```

OUTPUT

```
> python3 experiment_05.py
------------------------------------------------------------------------------------------------
Original Word    | Stemmed Output  | Suffix Stripped
-------------------------------------------------
connect          | connect         |
connection       | connect         | ion
connected        | connect         | ed
connecting       | connect         | ing
connective       | connect         | ive
program          | program         |
programs         | program         | s
programmer       | programm        | er
programming      | program         | ming
programmed       | program         | med
compute          | comput          | e
computation      | comput          | ation
computing        | comput          | ing
computer         | comput          | er
computable       | comput          | able
generous         | gener           | ous
generosity       | generos         | ity
revival          | reviv           | al
allowance        | allow           | ance
rational         | ration          | al
[Process completed with exit code 0]
```

RESULT

Thus, morphological word stemming using the Porter Stemmer algorithm was successfully implemented and its rule-based affix stripping verified.